

EDUCATION

Kent State University | BS. Computer Science

May 2025/GPA 3.75

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Network Security, Network Systems, Software Engineering, Secure Programming, Artificial Intelligence, Machine Learning, Human Interface Computing, Big Data

SKILLS

Languages: Python, C++, JavaScript/TypeScript, HTML/CSS, C#, .Net

AI/ML: TensorFlow, Scikit-learn, Pytorch, Numpy, Pandas, RAG

Web Development: React, Next.js, Node, GraphQL, Django, RESTful APIs, gRPC, WordPress

Mobile Development: Android Studio, React Native

Cloud and Tools: MySQL, PostgreSQL, AWS, Azure, Git, Postman, GitHub, Figma, Jira, Docker

EXPERIENCE

Gao Tek Inc. | Software Engineer Intern

Sept 2025 - Present

- Engineered advanced, plugin-free functionality with PHP and JS, including an AJAX-based live search feature and a custom module, and incorporated an AI chat agent, tuned to assist specific users.
- Architected an e-commerce platform and engineered bulk product data imports via CSV files.
- Mentoring 73 interns in debugging, fixing errors in SEO, structure, responsiveness, and UI functionality.

Blockchain & ML Research Assistant

Sept 2024 - May 2025

- Co-led a research project to architect a secure on-chain data layer for AI models, mapping data flows and reviewing consensus protocols (PoW, PoS) to ensure data accuracy, integrity, and auditability.
- Designed use cases for blockchain-verified datasets in TinyML training pipelines, detailing how immutable provenance and integrity checks can boost query precision and reduce inference overhead.
- Authored a review paper on advanced reasoning-capable LLMs, analyzed pre-training methods, RLHF and distillation pipelines, CoT prompting, and RL optimizers.

PROJECTS

[Machine & Deep Learning](#) | Model Developer | Fine-Tuning

Spring 2025

- Headed end-to-end construction of a predictive ML pipeline to predict student performance based on country, topic, sub-topic, and associated keywords, using a dataset of 9546 records.
- Formulated, trained, and examined two supervised models: a TensorFlow deep learning sequential model (60% accuracy) and a Scikit-learn XGBoost classifier (72.35% accuracy)

[Facial Expression Recognition](#) | Model Developer | Fine-Tuning

Fall 2024

- Built and trained a deep Convolutional Neural Network (CNN) using TensorFlow and Keras to classify 7 distinct human facial expressions (e.g., Happy, Sad, Angry) from a public Kaggle dataset.
- Engineered the model architecture with multiple convolutional blocks, BatchNormalization, and GlobalAveragePooling2D to improve feature extraction and reduce parameters.
- Improved model accuracy from 55% to 58% by optimizing and preventing overfitting through data splitting, L2 regularization, Dropout, and EarlyStopping callbacks.

[Image Explainer](#) | GenAI Mobile-App Developer

March 2024

- Spearheaded delivery of a cross-platform React Native app to enable real-time object recognition for visually impaired users; integrated Gemini Gen AI Vision & Language APIs via React Native's fetch client and invoked native Text-to-Speech bridges for instant audio feedback.
- Engineered a Node.js backend with 5+ RESTful JSON endpoints, implementing JWT-based stateless authentication, bcrypt password hashing, and secure session tokens to secure user accounts.
- Designed and optimized a relational SQL database with normalized schemas, indexed columns, and connection pooling, resulting in a 19% reduction in average query latency

[NEO Volleyball](#) | Full Stack Developer

Fall 2024

- Built a cross-platform tournament manager using React.js (Bootstrap) and Django REST Framework.
- Developed RESTful APIs, JSON parsing, and custom middleware for real-time frontend-backend sync for reactive state management, client/server-side form validation, and live UI shuffling of players.
- Automated Excel-based processes by creating import/export pipelines and dynamic form generators; implemented modular components to support concurrent edits and live data sync for 2500 users.
- Built JWT-based auth system, registration portal & calendar scheduler with React Hooks and Context API. Designed UML diagrams, system architecture docs, & showcased the app as a final presentation.